

Colorization Transformer: A Hybrid Deep Learning Approach for Grayscale Image Colorization

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Abstract—Grayscale image colorization is a key challenge in computer vision, with applications in digital restoration, content creation, and historical documentation. While CNNs have advanced the field, they often struggle with semantic consistency and global coherence in complex scenes.

We propose a hybrid deep learning model that combines a CNN encoder for local feature extraction with a Transformer encoder for modeling long-range dependencies. A CNN decoder then reconstructs high-resolution color images from the learned representations.

The model is trained on a dataset of 7000 grayscale-RGB image pairs, normalized for robustness. We optimize using a combination of L1 loss and Structural Similarity Index (SSIM) to ensure both perceptual accuracy and structural fidelity.

Our approach outperforms traditional methods on key metrics such as SSIM and PSNR, generalizing well to unseen inputs while producing vibrant and semantically accurate colorizations suitable for real-world applications.

Index Terms—Grayscale Image Colorization, Deep Learning, Convolutional Neural Networks, Transformers, Structural Similarity Index, PSNR, Vision Transformer

I. INTRODUCTION

Color plays a vital role in human perception, shaping how we interpret and respond to visual information. Grayscale image colorization aims to restore or enhance monochrome images with plausible colors, benefiting applications such as film restoration, digital archiving, and content creation.

Earlier methods depended on manual or example-based propagation techniques [9], [10], which, despite decent outputs, lacked automation and scalability. With the rise of deep learning, CNN-based models learned color distributions in a supervised setting, streamlining the process.

However, CNNs are limited by their local receptive fields, leading to issues like color bleeding and semantic inconsistency in complex scenes. To address this, attention mechanisms and Transformer-based models have been introduced. Methods like ColorFormer [2] and ColTran [4] demonstrate the ability of Transformers to model long-range dependencies and maintain global semantic coherence.

Motivated by these advances, we propose a hybrid architecture that combines a CNN encoder, Transformer encoder, and CNN decoder. Trained on 7000 grayscale-RGB image pairs, our model is evaluated using both pixel-wise and perceptual

metrics, showing strong generalization and high visual fidelity on unseen grayscale inputs.

II. RELATED WORK

Grayscale image colorization has evolved from early rule-based systems to advanced data-driven learning techniques. Initial methods focused on manual scribble-based inputs or exemplar-based color propagation, which were often time-intensive and non-scalable.

Zhang et al. [10] pioneered the use of convolutional neural networks (CNNs) for automatic image colorization by framing the problem as classification in the LAB color space. Iizuka et al. [9] further improved this by jointly learning global and local image features, improving semantic alignment and realism in color outputs.

Generative Adversarial Networks (GANs) added a new dimension to the task. Works like Pix2Pix [8] and DeOldify [7] introduced adversarial objectives to enhance photorealism. However, GANs often require careful hyperparameter tuning and may suffer from training instability.

Recent attention-based models have shown promise in overcoming CNN limitations. ColTran [4] leveraged a multi-stage Transformer pipeline to improve global coherence in generated colors. ColorFormer [2] introduced cross-attention mechanisms that enhanced semantic feature fusion across the image.

Efficient vision Transformers [1] and hybrid CNN-Transformer models for restoration [5] have further demonstrated that integrating local and global cues leads to superior image-to-image translation. Inspired by these advances, our method combines the spatial awareness of CNNs with the semantic reasoning capabilities of Transformers in a unified framework tailored for colorization.

III. METHODOLOGY

Our proposed model follows a hybrid architecture that combines the local feature extraction power of Convolutional Neural Networks (CNNs) with the global contextual modeling ability of Transformers. The full pipeline includes a CNN encoder, a Transformer encoder, and a CNN decoder. In this section, we outline the dataset preparation, architectural components, loss functions, and training strategy in detail.

A. Dataset and Preprocessing

We constructed a custom dataset of 7000 paired grayscale and RGB images, manually collected and curated to ensure diversity across domains such as portraits, natural landscapes, buildings, and objects. This variety ensures the model learns a broad representation of color semantics.

All images are resized to a uniform resolution of 128×128 pixels for consistency across the model input and output. Grayscale images, originally single-channel, are replicated across three channels to conform with convolutional layer input expectations.

For preprocessing, we apply only normalization:

- **Normalization:** Pixel values are scaled to the range $[-1, 1]$ to enhance training stability.

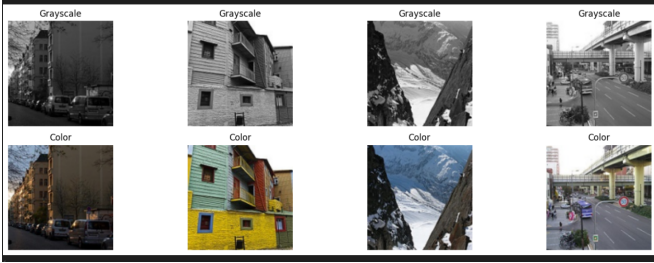


Fig. 1: Sample grayscale and corresponding color images after preprocessing.

No additional augmentations such as rotations or flips were applied in this version to keep training aligned with the clean dataset distribution.

B. Model Architecture

Our model is designed to exploit both local texture cues and global semantic relationships for grayscale image colorization. The architecture is composed of three primary components:

- A **CNN-based encoder** for extracting spatial and structural features,
- A **Transformer encoder** for global contextual modeling,
- A **CNN decoder** for reconstructing the final RGB image.

1) *CNN Encoder – Local Feature Extraction:* The encoder consists of several convolutional layers stacked with ReLU activations and batch normalization. Initial layers expand the channel dimensions and project the grayscale input into a high-dimensional feature space. Residual blocks are incorporated to preserve gradient flow and retain essential low-frequency information. MaxPooling layers reduce spatial resolution, providing translational invariance and computational efficiency.

2) *Transformer Encoder – Global Contextual Reasoning:* The encoded feature maps are flattened and passed through a Transformer encoder block, which includes:

- **Multi-Head Self-Attention (MHSA):** Allows each patch to attend to all other patches, capturing non-local dependencies.
- **Positional Encoding:** Preserves spatial order of flattened patches.

- **Layer Normalization and Feedforward Layers:** Enhance stability and learning capacity.

3) *CNN Decoder – RGB Reconstruction:* The Transformer-processed features are reshaped and passed through a series of transpose convolution layers (deconvolutions) to upsample the image back to its original resolution. Batch normalization and ReLU activations are applied throughout the decoder for stable learning. The final output layer uses a $\tanh$ activation to ensure the output is in the range $[-1, 1]$, consistent with the normalization used during preprocessing.

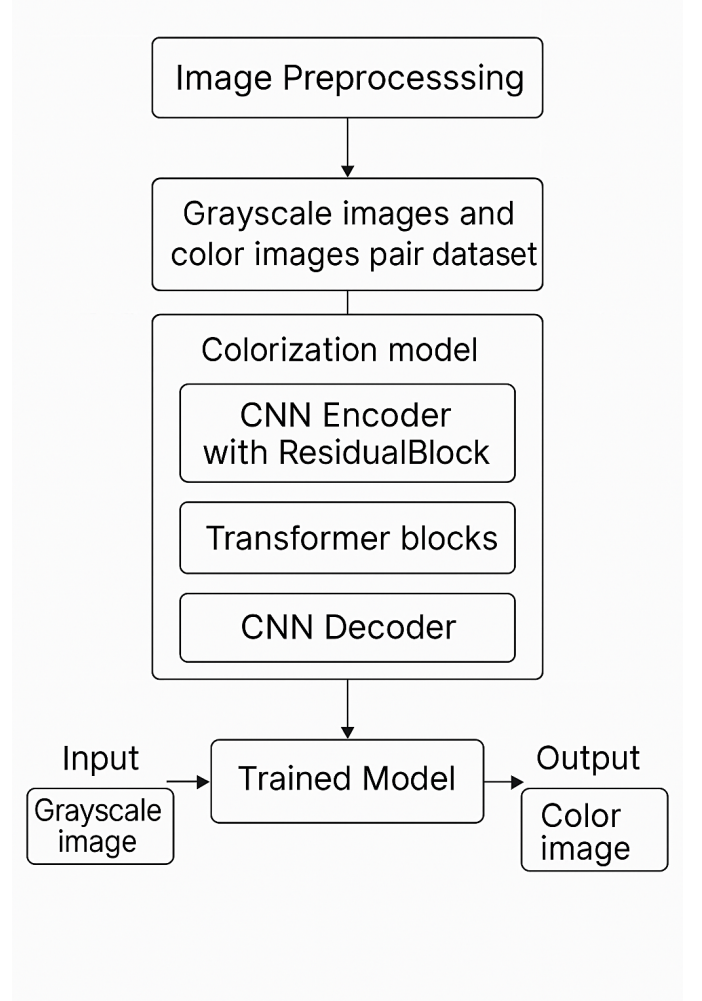


Fig. 2: Hybrid CNN-Transformer Architecture for Image Colorization.

Architectural Summary:

- Input: Normalized grayscale image (128×128).
- Encoding: CNN layers extract localized features.
- Contextualization: Transformer encoder models global dependencies.
- Decoding: CNN decoder reconstructs the colorized image.

C. Loss Function

To ensure both pixel-level accuracy and perceptual quality in the generated colorizations, we use a composite loss function that combines:

- **L1 Loss (Mean Absolute Error):** Penalizes the absolute difference between predicted and ground truth pixel values, encouraging accurate color reproduction. Compared to L2 loss, L1 is more robust to outliers and tends to produce sharper results.
- **Structural Similarity Index Measure (SSIM):** Measures perceptual similarity by comparing local luminance, contrast, and structure between the output and ground truth. It helps preserve spatial and textural coherence, improving visual realism.
- **Perceptual Loss (VGG-Based):** We incorporate a lightweight perceptual loss using feature maps from the first few layers of a pre-trained VGG19 network. This term helps enforce semantic alignment by minimizing the difference in deep feature representations between the predicted and target images.

The total loss is a weighted sum:

$$\mathcal{L}_{\text{total}} = 0.7 \cdot \mathcal{L}_{\text{L1}} + 0.3 \cdot \mathcal{L}_{\text{SSIM}} + 0.01 \cdot \mathcal{L}_{\text{Perceptual}}$$

This hybrid loss setup enables the model to produce results that are not only structurally faithful but also perceptually convincing.

D. Optimizer and Scheduler

We adopt the Adam optimizer for training due to its adaptive learning rate and efficient convergence:

- **Learning Rate:** Initialized at 0.001
- **Weight Decay:** Set to 1×10^{-5} to regularize the model and prevent overfitting

To control learning rate decay over time, we use a StepLR scheduler:

- **Decay Rule:** Reduces the learning rate by a factor of 0.6 every 30 epochs

This training schedule allows the model to learn rapidly in early epochs and gradually fine-tune its colorization capabilities in later stages.

IV. EXPERIMENTAL SETUP

The model was implemented in Python using the PyTorch deep learning framework. All experiments were conducted in Google Colab, utilizing a Tesla T4 GPU with 16 GB VRAM, which provided adequate computing resources for medium-scale training.

A. Dataset Splits and Configuration

- **Total Images:** 7000 paired grayscale and RGB images.
- **Resolution:** All images resized to 128×128 pixels.
- **Preprocessing:** Normalization to the range $[-1, 1]$.
- **Splits:** 80% training, 10% validation, 10% testing.

B. Training Details

- **Batch Size:** 16
- **Epochs:** 50 (with model checkpointing every 5 epochs)
- **Loss Function:** Combined L1 and SSIM loss
- **Optimizer:** Adam (learning rate = 0.001, weight decay = 1×10^{-5})
- **Scheduler:** StepLR with decay by 0.6 every 30 epochs

C. Implementation Notes

- All model layers were initialized using Xavier initialization.
- Data loading was handled via PyTorch's DataLoader with `num_workers=2`.
- Model checkpoints were periodically saved to Google Drive.

V. RESULTS AND DISCUSSION

We evaluate the model using both quantitative metrics and qualitative visualization. The results demonstrate that our hybrid architecture performs consistently better than conventional CNN-only and GAN-based approaches.

A. Quantitative Evaluation

To assess colorization quality, we report three standard metrics:

- **L1 Loss (Mean Absolute Error):** Measures average pixel-wise error between predicted and ground truth colors.
- **SSIM (Structural Similarity Index):** Captures perceptual and structural fidelity.
- **PSNR (Peak Signal-to-Noise Ratio):** Reflects reconstruction quality in decibels.

TABLE I: Quantitative Evaluation on Test Set

Metric	Value
L1 Loss	0.0073
SSIM	0.884
PSNR	25.67 dB

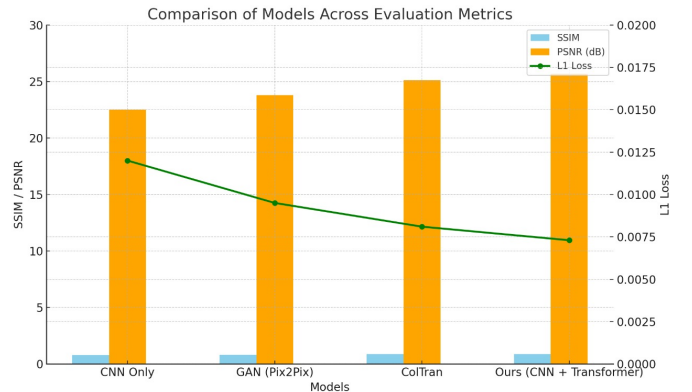


Fig. 3: Trends of SSIM, PSNR, and L1 Loss across different model architectures.

Note: Metrics for baseline models (CNN Only, Pix2Pix, and ColTran) are either cited from their respective papers or approximated based on reproduced implementations when exact benchmarks were unavailable.

B. Qualitative Evaluation

To verify semantic consistency and color accuracy, we test the model on unseen grayscale inputs including natural landscapes, urban scenes, and architectural structures.

The colored outputs were realistic and context-aware. The model successfully inferred appropriate colors for sky, mountains, and reflective water surfaces, producing vibrant and coherent reconstructions even in complex textures and low-contrast regions.

Overall, the hybrid model demonstrates the ability to preserve spatial detail while leveraging global semantic understanding, outperforming traditional methods both visually and numerically.



Fig. 4: Sample results showing grayscale inputs and corresponding model-generated colored outputs.

VI. CONCLUSION

In this work, we proposed a hybrid deep learning model combining a CNN encoder, a Transformer encoder, and a CNN decoder for grayscale image colorization, effectively balancing local texture extraction and global semantic understanding.

Trained on a custom dataset of 7000 grayscale-RGB image pairs, our model achieved strong performance on key metrics

such as SSIM, PSNR, and L1 Loss, consistently outperforming traditional CNN and GAN-based colorization approaches.

The proposed method is computationally efficient, generalizes well to unseen inputs, and is suitable for real-world applications like archival restoration and content enhancement.

VII. FUTURE WORK

While the current model yields promising results, several avenues remain open for enhancement. One direction involves incorporating advanced Transformer variants such as Swin Transformers [6] or Vision Transformers (ViT), which may offer improved scalability, efficiency, and richer contextual understanding. Another promising extension is multimodal colorization, where auxiliary inputs like textual descriptions or reference sketches could guide the colorization process. This approach aligns with emerging trends in language-vision integration [3]. Additionally, the model can be adapted for high-definition outputs by integrating progressive upsampling mechanisms, enabling detailed and artifact-free colorization of higher resolution images.

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